

Arjuna JEE 2025

Maths

DPP: 8

Basic Mathematics

Q1 Suppose $abc > 0$ and $a + b + c < 0$ and $\frac{|a|}{a} + \frac{|b|}{b} + \frac{|c|}{c} = x$ then value of $x^3 - 6x^2 + 11x - 6$ equals

- (A) 0 (B) -24
(C) 12 (D) 10

Q2 If $f(x) = |2 - x| + |4 - x|$ and $\frac{f(\pi) + f(2.1)}{f(3)} = \lambda$ then the value of $\frac{17\lambda}{4}$ is

- (A) 8.5
(B) 8
(C) 4.25
(D) 17

Q3 $x^2 + |x - 1| = 1$

Q4 Solve $\left| \frac{x-3}{x+1} \right| \leq 1$

- (A) $x > -1$ (B) $x \leq -1$
(C) $x \geq 1$ (D) $x > -1$

Q5 Solve $|2^x - 1| + |2^x + 1| = 2$

- (A) $x \in (-\infty, 0]$
(B) $x \in (-\infty, 1]$
(C) $x \in [0, \infty)$
(D) $x \in [1, \infty)$

Q6 Solve for x :

- (A) $|x| > 5$

(B) $|x - 1| \leq 3$

(C) $|2x + 3| \geq 4$

Q7 If $\sqrt{x^2 - 6x + 9} + \sqrt{e^y - 1} + \sqrt{\tan^2 \theta - 3} = 0$ then the value of $x + y + \sec^2 \theta$ is

Q8 If $b = \sqrt{6 + 2\sqrt{5}} + \sqrt{6 - 2\sqrt{5}}$, then the value of b is equal to

- (A) $3\sqrt{5}$
(B) $2\sqrt{5}$
(C) $6\sqrt{3}$
(D) $8\sqrt{2}$

Q9 Sum of all the roots of $x^2 - |x| - 6 = 0$ is

- (A) 0 (B) 6
(C) 3 (D) 5

Q10 Solve the following equation: $|x|^3 - 7|x| + 6 = 0$

Q11 Solve the following equation: $|x| - |x - 2| = 2$

Q12 Solve the following equation: $|x - 3| + 2|x + 1| = 4$

Q13 Solve the following equation: $\left| \frac{x^2 - 5x + 4}{x^2 - 4} \right| = 1$

Q14 Solve the following inequalities: $|5 - 2x| < 1$

Q15 Solve the following inequalities: $|6x - 5| \geq 4$



Answer Key

Q1 (B)

Q2 (A)

Q3 $x=0,1$

Q4 (C)

Q5 (A)

Q6 A) $x \in (-\infty, -5) \cup (5, \infty)$

B) $x \in [-2, 4]$

C) $x \in (-\infty, -3.5) \cup (0.5, \infty)$

Q7 7

Q8 (B)

Q9 (A)

Q10 $\pm 1, \pm 2$

Q11 $x \in [2, \infty)$

Q12 $x = -1$

Q13 $x = 0, 8/5, 5/2$

Q14 $x \in (2, 3)$

Q15 $x \in (-\infty, 1/6] \cup [3/2, \infty)$



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